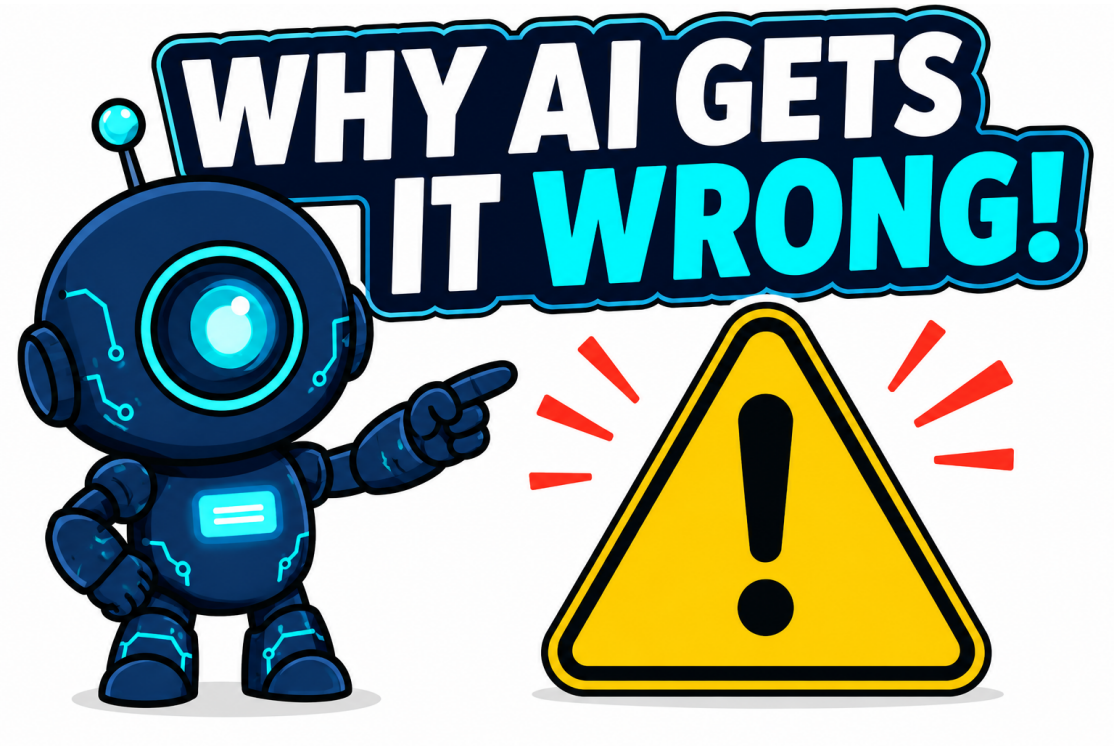
AI MECH LAB

AI Course — Beginner Track



Module 1, Topic 1.3

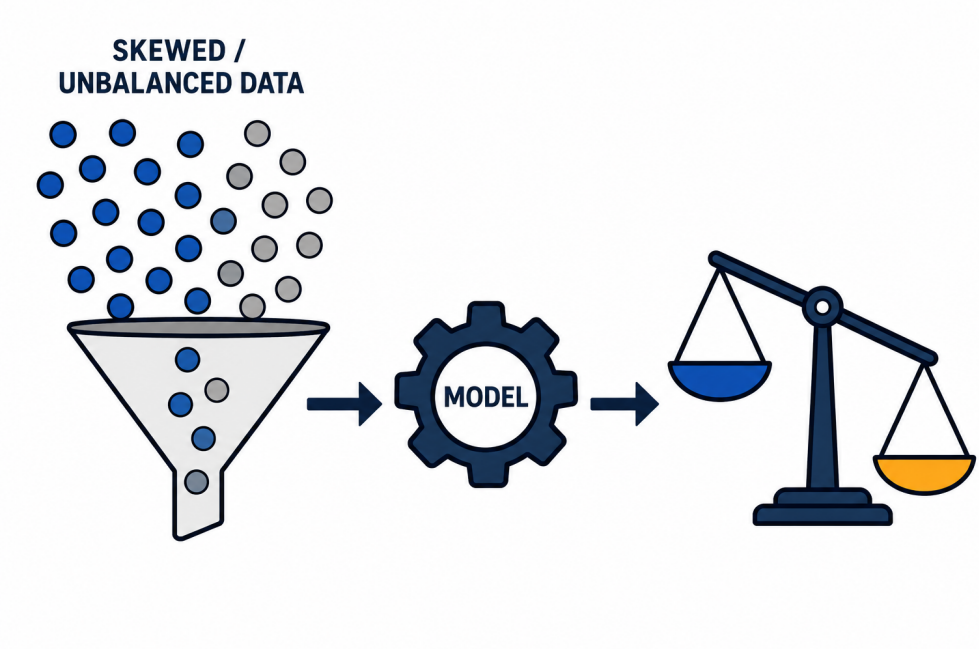
AI Ethics and Limitations

Lesson

You now know what AI, Machine Learning, and Deep Learning are, and you've seen real examples of them working well. This lesson covers the other side: what happens when these systems fail, why they fail in ways that are easy to miss, and what that means for you as someone who'll eventually build or deploy them.

Why bias isn't a bug in the traditional sense

When a rule-based system fails, it's usually because a programmer wrote a rule wrong — a bug you can find and fix by reading the code. Machine learning systems fail differently. They learn patterns from historical data, and if that data reflects a historical bias, the model doesn't correct for it — it learns the bias as if it were a legitimate pattern to replicate.



Skewed or unbalanced training data produces a skewed model, and a skewed model produces unbalanced outcomes.

Amazon found this out the hard way. It built an internal AI tool to screen job resumes, trained on ten years of past submissions — a period when the tech industry's applicant pool skewed heavily male. The model concluded, from the data, that male candidates were preferable. It began penalizing resumes containing the word "women's" — even in an unrelated context like "women's chess club captain" — and downgraded graduates of women's colleges. Nobody programmed the tool to discriminate. It learned to, because the historical data it was trained on already reflected that pattern, and the model's job was to find and replicate patterns.

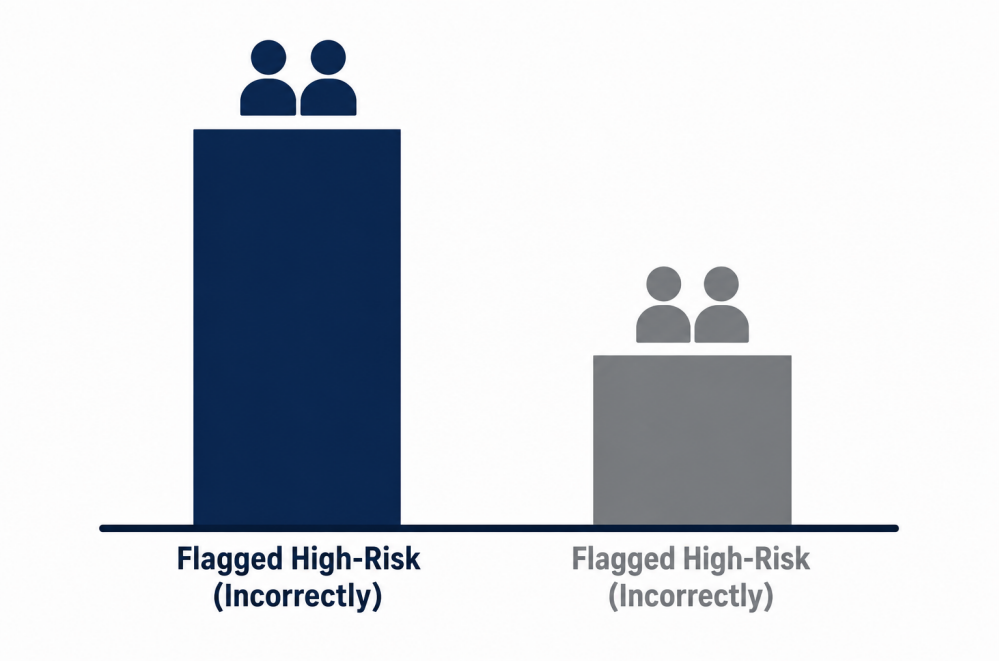


An AI resume screener penalizing a qualified candidate based on a learned historical pattern, not an explicit rule.

This is the core lesson: an ML system trained on biased historical data will learn and often amplify that bias, even with no explicit intent from anyone involved in building it.

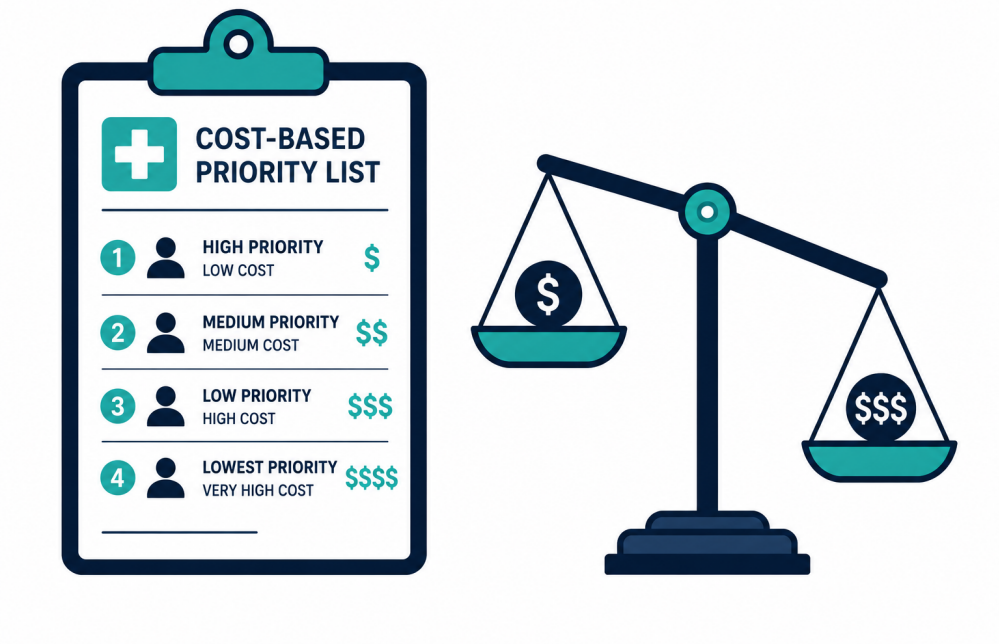
When the stakes are higher than hiring

The COMPAS algorithm, used in parts of the US court system to assess a defendant's risk of reoffending, was investigated by the journalism organization ProPublica and found to produce racially disparate outcomes — Black defendants were flagged as high-risk at nearly twice the rate of white defendants who did not reoffend, while white defendants who did reoffend were more often incorrectly scored as low-risk. The historical arrest and sentencing data used to train the model reflected existing disparities in the justice system, and the model reproduced them in its predictions — predictions that then influenced real sentencing and bail decisions.



Disparity in incorrect high-risk flags produced by the COMPAS algorithm.

A similar pattern showed up in a healthcare algorithm used across US hospitals to decide which patients needed extra care management. The model used past healthcare spending as a stand-in for how sick a patient was — a reasonable-sounding assumption on the surface. But because Black patients had historically received less healthcare spending due to unequal access to care (not because they were healthier), the algorithm concluded they needed less follow-up care than white patients with comparably serious conditions, at scale, across over 200 million patients.



A cost-based priority system: using healthcare spending as a proxy for medical need.

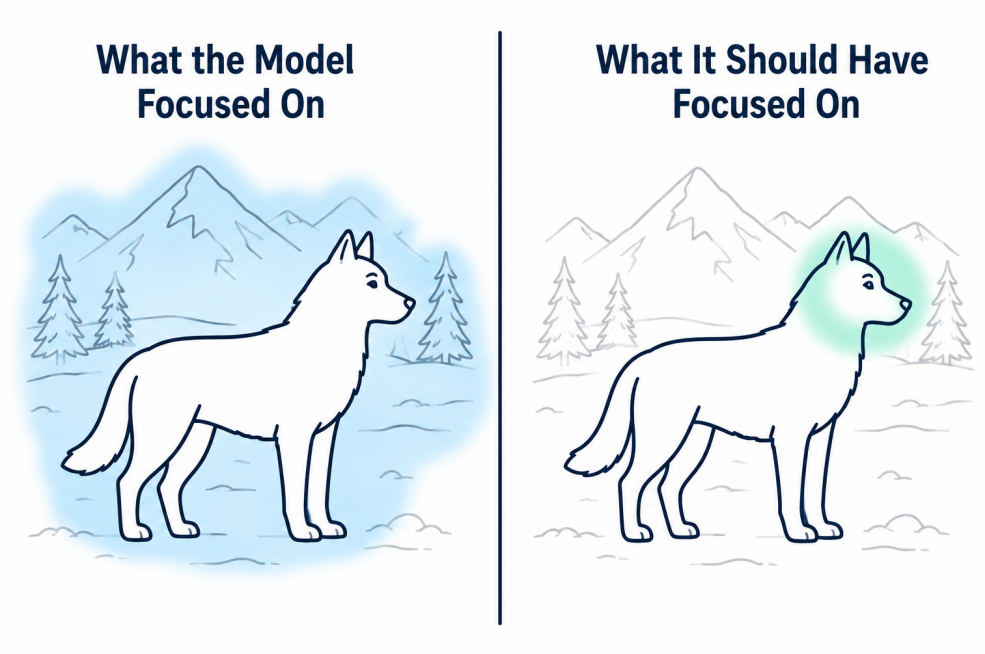
Notice the common thread across all three cases: none of these systems were fed an explicit rule to discriminate. They were fed historical data that already contained inequality, and they learned that

inequality as a pattern worth repeating.

When AI learns the wrong pattern entirely

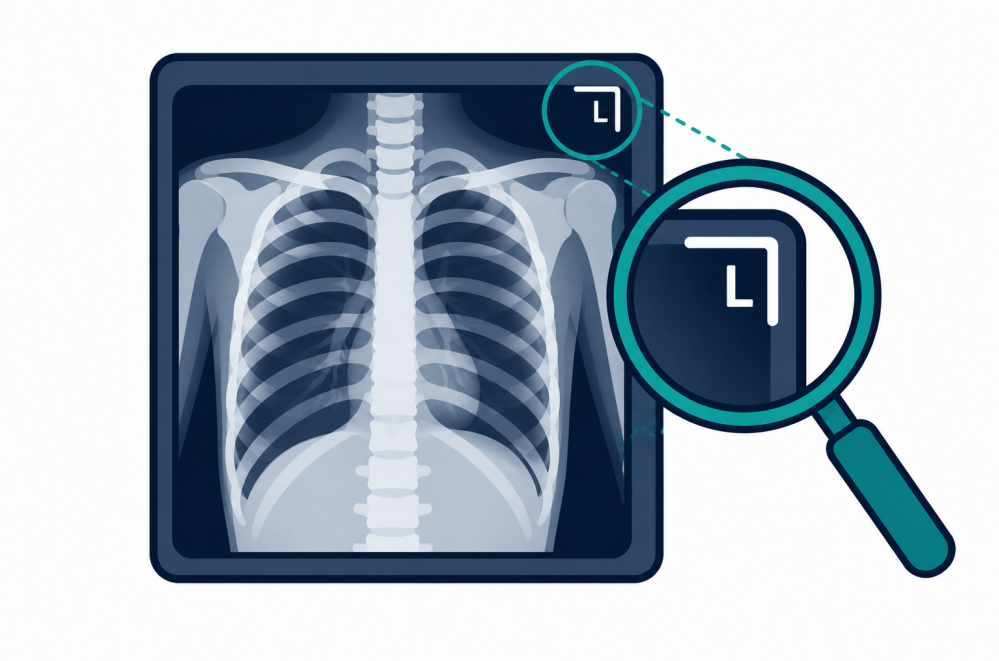
Bias from historical data is one failure mode. A separate, equally important one is when a model learns to key off something that isn't actually relevant to the problem at all — and looks accurate until it's tested somewhere new.

A well-known computer vision study trained a model to tell wolves apart from huskies. It performed close to perfectly on its test data. But when researchers dug into why it was making its decisions, using a technique that visualizes which parts of an image the model is actually paying attention to, they found something unexpected: the model wasn't looking at the animals at all. It was checking whether the image background contained snow — because every wolf photo in its training set happened to be shot in a snowy setting. The model had learned "snow equals wolf," not "these facial features equal wolf." It got the right answer on its test set by complete coincidence of how that dataset happened to be composed.



What the model actually paid attention to, versus what it should have focused on.

A pneumonia-detection model trained on hospital X-rays ran into a version of the same problem. It performed at expert level in initial testing, then failed when used at a different hospital. It turned out the model had learned to recognize the specific style of metal markers and text fonts that different X-ray machines stamped onto images — and since certain machines were used more often in high-risk ICU settings than in routine outpatient clinics, the model was effectively predicting "which room was this X-ray taken in," not "what does this patient's lung tissue actually look like."

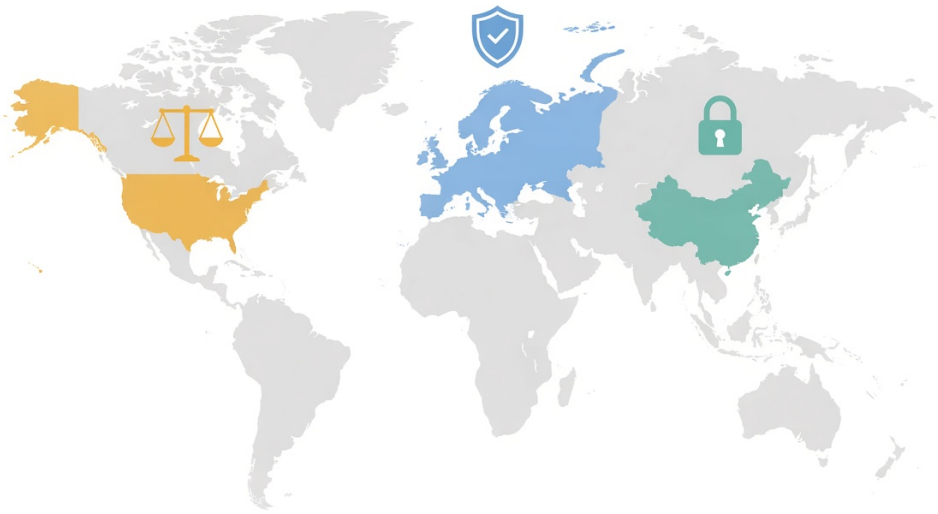


A model trained to key off a machine's corner marker instead of the actual lung tissue.

This is the practical lesson to carry forward: a model that performs well on its test data is not the same as a model that has learned the thing you actually wanted it to learn. It may have found a shortcut — a pattern that correlates with the right answer in your specific dataset, without being the actual cause of that answer. This is one of the most important and most under-appreciated limitations of machine learning, and it's a large part of why deep learning systems, in particular, need careful testing on genuinely new data before anyone trusts them in a real application.

Where regulation currently stands

Because failures like these have real consequences, governments have started building rules around how AI can be used, though there's no single global standard. Three different approaches currently coexist: the European Union has passed a legally binding, risk-based framework that places strict requirements — mandatory oversight, auditing, transparency rules — on "high-risk" AI uses like hiring and credit scoring. The United States has taken a more decentralized approach, relying on existing sector regulators, state-level rules, and voluntary frameworks rather than one sweeping federal law. China has taken a centralized approach focused heavily on content control for generative AI, requiring registration and security review for publicly released models.

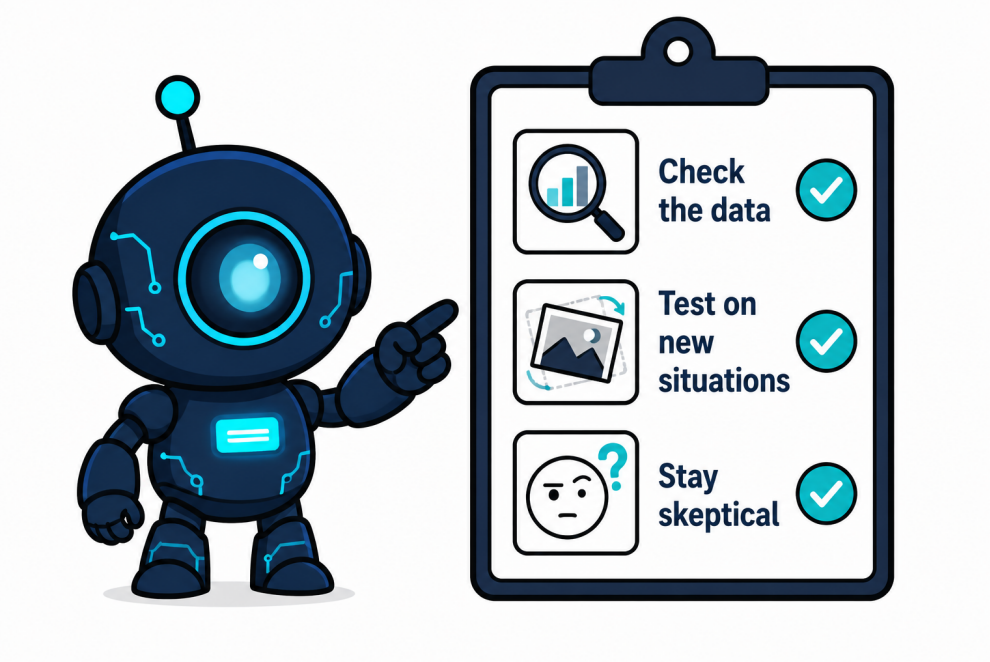


Three distinct regulatory approaches: the EU, the United States, and China.

This is a fast-moving area, and the specific rules are likely to keep shifting — the pattern worth remembering is that meaningful oversight now exists in some form across major markets, which didn't used to be true.

Where this leaves you

The takeaway isn't that AI is untrustworthy — it's that AI is only as good as the data and testing behind it, and it fails in ways that are often invisible until someone specifically looks for them. As you move further into this course and start building your own models, this lesson is the one to keep returning to: check what your data actually represents, test on genuinely new situations, and stay skeptical of a model that looks too accurate too easily.



Three habits to carry forward: check the data, test on new situations, stay skeptical.